Corporate bond issuer-bondholder legal structure?

Indenture – contract between issuer and bondholders

Corporate trustee – bank/trust co. acting as fiduciary for investors

- Paid by issuer
- Enforces B/H interests in indenture
- Can declare default if issuer breaches indenture

Corporate bond fundamentals?

Source: HFIS Ch. 10: Corporate Bonds

- **Issuers:** utilities, transportation, industrial, banks, etc.
- **Maturity:** typically < 30 years
- Most are **registered** or **book entry** → automatic coupons
- **Straight coupon bonds:** fixed coupons paid semiannually
  - Participating bonds share profits above certain level
  - Income bonds (rare) pay interest only if issuer income sufficient
- Coupons may also be zero, deferred, or paid-in-kind (PIK)

Compare Zero vs. Deferred vs. PIK coupons.

Source: HFIS Ch. 10: Corporate Bonds

- **Zero-coupon bonds**
  - A type of OID bond
  - Less popular today (tax benefits declined)
  - No reinvestment risk if held to maturity
- **Deferred-interest bond (DIB)** – blend of straight coupon + zero coupon
  - Most are junk bonds
  - Most pay no cash interest for first 5 years
- **Pay-in-kind (PIK) debentures**
  - Pays additional securities instead of coupons
  - Additional securities can be sold separately

Note: All have **higher credit risk** than straight coupon bonds

Describe 5 bond security types.

Source: HFIS Ch. 10: Corporate Bonds

- **Debentures** – no specific security (most common)
  - B/H = general creditors; security = issuer financial strength
  - Common restrictions: min net worth, selling major assets, stock dividends
  - Other B/H protections: negative pledge clause
  - Subordinated issues may offer conversion rights
- **Mortgage Bonds** – B/H gets 1st-mortgage lien on property
  - Blanket mortgage → allows additional series on same mortgage
  - Possible B/H protection: after-acquired clause
- **Collateral Trust Bonds** – backed by stock of issuer's subsidiaries
  - B/H protection: issuer default → forfeits voting rights on subsidiary stock
- **Equipment Trust Certificates** – issued by railroads
  - Trustee owns RR equipment, leases to RR, RR buys at lease end
  - Very secure: RR depends on equipment, always buys back
- **Guaranteed Bonds** – guaranteed by one or more other companies

List 5 early debt retirement methods.

Source: HFIS Ch. 10: Corporate Bonds

1. **Call and refunding provisions**
2. **Sinking fund provisions**
3. **Maintenance and replacement funds** (doesn't retire bonds)
4. **Redemption through sale of assets** (usually restricted)
5. **Tender offers** – issuer buys back bonds at PV using CMT + fixed spread

First 4 must be in indenture if applicable

Describe corporate bond call provisions.

- **Riskiest for B/H: fixed price call**
  - Call price usually starts at premium, then declines
  - May have initial lockout period or non-refundable period
- **Less risky, more common: make-whole call**
  - Call price = max(par, PV remaining CFs at CMT + spread)
  - Call price moves inversely with interest rates
  - Issuer advantage: can issue at lower yield

Describe corporate bond sinking fund provisions.

Sinking fund provisions popularity has declined

- Indenture requires issuer to retire portion of bonds each year (usually at par)
- $\downarrow$ default risk
- B/H benefit if rates $\uparrow$ (get par even if MV $<$ par)
- Issuer may accelerate sinking fund if rates $\downarrow$

Issuer default rate formula?

Issuer default rate – based on number of defaulting issuers

$$\frac{\text{No. Issuers that Default in Year}}{\text{Total Issuers at BOY}}$$

Dollar default rate formula?

Dollar default rate – based on amount of defaulted par

Over multiple years:

$$\frac{\text{Cumulative \$ Value of Defaulted Bonds}}{\text{Cumulative \$ Value of All Issues} \times \text{Wtd Avg Years Outstanding}}$$

Annual rate:

$$\frac{\text{Cumulative \$ Value of Defaulted Bonds}}{\text{Cumulative \$ Value of All Issues}}$$

Default loss rate and recovery rate formulas?

Source: HFIS Ch. 10: Corporate Bonds

Default Loss Rate = Default Rate $-$ Recovery Rate

Recovery rate = % of defaulted bond B/H recover after default

- Difficult to measure: consists of cash and securities

Event risk vs. headline risk?

Source: HFIS Ch. 10: Corporate Bonds

Event risk – stockholder-driven event → lower bond prices

- Rating downgrades, ↑ leverage from M&A, etc.

Headline risk – bad media coverage → lower bond prices

Describe high-yield bond unique features.

Source: HFIS Ch. 10: Corporate Bonds

A.k.a. **"junk bonds"** – many only just below IG ($<$ BBB or Baa)

- **Deferred Coupon Structures** (3 kinds)
  1. **Deferred-interest bonds** (most common) – no coupons 3–7 years
  2. **Step-up bonds** – low initial coupon, then ↑
  3. **PIK bonds**
- **Extendible Reset Bonds**
  - Coupon resets periodically to maintain specified price
  - New rate = new interest rate level + credit spread
  - $\neq$ floating rate (which uses fixed spread)
  - In practice, issuer may not afford the coupon
- **Clawback** – redeem bonds during non-callable period w/ IPO proceeds
  - Can hurt B/H since investment would be rising

Mortgage payment characteristics?

Source: HFIS Ch. 21: Mortgages

- Can be amortizing (principal and interest), IO, and/or balloon
- Rate can be fixed, adjustable (ARMs), or hybrid ARMs
  - ARMs require payment to be recast when the index-based rate changes

Key mortgage industry participants?

Source: HFIS Ch. 21: Mortgages

- **Direct lenders** – UW and provide loans directly to borrowers
- **Brokers** – work with clients and **wholesale lenders**
- **TPOs** – when a loan is completely/partially originated/processed by a party other than seller
- **Depository institutions** – fund loans with bank deposits
- **Non-depository lenders** – fund loans by selling them to investors
- **Originators** – UW and fund loans
- **Servicers** – collect monthly payments, handle property tax payments, deal with delinquencies
- **MBS providers** – public or private companies that buy and securitize mortgages for investors

Describe aspects of the mortgage underwriting process.

Source: HFIS Ch. 21: Mortgages

UW evaluates borrower's credit and value of property

Criteria used to evaluate credit:

1. Credit ("FICO") scores: 730+ is strong
2. If LTV ratio $>$ 80%, mortgage insurance required
3. Debt-to-income ratio criteria (lower is better) (e.g. front ratio and back ratio)

$$\text{Front Ratio} = \frac{\text{Total Monthly Payments on the Home}}{\text{Borrower's Pretax Monthly Income}}$$

$$\text{Back Ratio} = \text{Front Ratio} + \frac{\text{Other Debt Payments Like Auto Loans}}{\text{Borrower's Pretax Monthly Income}}$$

Documentation/standards $\downarrow$ until 2008 crisis, then $\uparrow$

How are mortgage lending rates generated?

Source: HFIS Ch. 21: Mortgages

- MBS providers heavily influence mortgage rates
- **WAC** = Weighted avg coupon in MBS pool
- Originators set rates $>$ MBS coupon to profit
- MBS investor coupon $\approx$ 25–75 bps $<$ WAC due to:
  1. Servicing fee
  2. Guaranty fees
  3. Excess servicing fee
- **Discount points** – % of loan upfront fee to borrowers
- **Subordination** – non-agency credit enhancement; junior bonds absorb losses first

How do rates affect prepayment/duration?

Source: HFIS Ch. 21: Mortgages

- Falling interest rates: prepayments $\uparrow$, durations $\downarrow$
- Rising interest rates: prepayments $\downarrow$, durations $\uparrow$ (“extension”)

List 5 factors impacting mortgage prepayment risk.

Source: HFIS Ch. 21: Mortgages

1. Sale of property
2. Destruction of property (e.g. fire)
3. Borrower default
4. Partial principal prepayments
5. Refinancing

Prepayments can be rate-sensitive or rate-insensitive

Define SMM, CPR, and the PSA model.

Source: HFIS Ch. 21: Mortgages

1. **SMM** (Single Monthly Mortality)

$$\text{SMM} = \frac{\text{Scheduled Bal} - \text{Actual Bal}}{\text{Scheduled Bal}}$$

2. **CPR** (Conditional Prepayment Rate) = annualized SMM

$$\text{CPR} = 1 - (1 - \text{SMM})^{12}$$

3. **PSA Model** – CPRs not constant over loan life
   - CPR $\uparrow$ by 0.2% per month until 6.0% in month 30
   - Can express in multiples

List 5 mortgage credit risk factors.

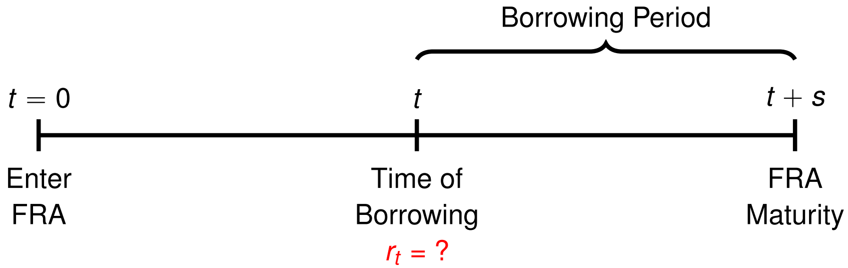
Source: HFIS Ch. 21: Mortgages

1. Credit scores
2. LTVs
3. Number of days delinquent
4. Default (90+ days delinq.)
   - Some agencies insure (Ginnie Mae)
   - Non-agency recoveries depend on foreclosure process
5. Default loss severity ↑ when:
   - High LTV
   - Appraisal values ¿ market values
   - Property values ↓ after origination
   - Foreclosure process is expensive

What is an FRA?

Source: ILA101-113 Ch. 7: Interest Rate Forwards and Futures

FRAs allow a borrower (lender) to lock in a borrowing (lending) rate for a future period



FRA payoff formula?

A borrower enters an FRA to hedge against rising borrowing costs

The borrower's payoff is positive if the actual borrowing rate $> r_{\text{FRA}}$

$$\text{Borrower's Payoff} = \begin{cases} (r_t - r_{\text{FRA}}) \times \text{Notional} & \text{if paid in arrears} \\ \dfrac{(r_t - r_{\text{FRA}})}{1 + r_t} \times \text{Notional} & \text{if paid at time of borrowing} \end{cases}$$

The lender's payoff is always the negative of the borrower's

If paid at time of borrowing, the payment is **"tailed"** by the current rate r_t

How to create synthetic FRA?

Let $r_{t,t+s}$ be the implied forward rate between time t and $t+s$

$$1 + r_{t,t+s} = \frac{P_t}{P_{t+s}}$$

where P_t and P_{t+s} are the current prices of zeros maturing at time t and $t+s$

To lock in $r_{t,t+s}$ as a borrowing rate:

1. Short $1 + r_{t,t+s}$ zero-coupon bonds maturing at time $t+s$
2. Buy 1 zero-coupon bond maturing at time t

Reverse the long/short position on each bond to lock in a lending rate

Eurodollar futures for hedging?

To lock in a borrowing (lending) rate, short (buy) Eurodollar futures

The payoff on a single short \$1M Eurodollar contract with futures price F_t is:

$$\begin{aligned}\text{Payoff} &= \left[F_t - (100 - \text{LIBOR}_t)\right] \times 100 \times 25 \\ &= \left[(100 - r_{\text{forward}}) - (100 - \text{LIBOR}_t)\right] \times 100 \times 25 \\ &= \left[\text{LIBOR}_t - r_{\text{forward}}\right] \times 100 \times 25\end{aligned}$$

The payoff is positive if LIBOR is higher than the locked-in forward rate

The lender's payoff is always the negative of the borrower's

Eurodollar futures vs. FRAs?

Source: ILA101-113 Ch. 7: Interest Rate Forwards and Futures

Both can be used to lock in borrowing or lending rates

Key difference: Eurodollar futures systematically favor borrowers

- The Eurodollar futures payoff is always at time of borrowing
- FRAs can payoff at time of borrowing or arrears
- If paid at time of borrowing, FRAs are automatically tailed at current rate
- Eurodollar futures can only be tailed at the forward rate (“convexity bias”)

$$\text{Payoff} = \frac{[F_t - (100 - \text{LIBOR}_t)] \times 100 \times 25}{1 + r_{\text{forward}}}$$

LIBOR vs. 3-Month T-Bills?

LIBOR $>$ Treasury rates because of:

1. Default premium in LIBOR rates
2. Higher liquidity premium in LIBOR rates
3. Market fluctuations that cause spreads to widen
4. State tax exemption of Treasuries

LIBOR is far more common for futures contracts

- Tracks private lending rates better than T-bills

Duration formulas?

$$\Delta B \text{ per 1\% change in } y = B \times \left[-\text{ModDur}(\Delta y) + \frac{1}{2}(\text{Convexity})(\Delta y)^2 \right]$$

$$B = \sum_{t=0}^{n} v^t CF_t$$

$$\text{MacDur} = \frac{\sum_{t=0}^{n} t v^t CF_t}{B}$$

$$\text{ModDur} = \text{MacDur} \times v$$

Alternative to modified duration: PVBP or “DV01” = $\frac{\text{ModDur}}{100}$

Convexity formula?

Source: ILA101-113 Ch. 7: Interest Rate Forwards and Futures

$$\Delta B \text{ per 1\% change in } y = B \times \left[-\text{ModDur}(\Delta y) + \frac{1}{2}(\text{Convexity})(\Delta y)^2 \right]$$

$$B = \sum_{t=0}^{n} v^t CF_t$$

$$\text{Convexity} = \frac{\sum_{t=0}^{n} t(t+1)v^{t+2}CF_t}{B}$$

Describe how a Treasury note futures contract works.

Source: ILA101-113 Ch. 7: Interest Rate Forwards and Futures

A Treasury note futures contract will specify:

- Underlying bond's YTM and maturity (e.g. 6% 10-year T-note)
- Acceptable maturity range of delivered note: e.g. 6.5–10 years
- Conversion factor: price of delivered note at underlying's YTM (e.g. 6%)

Invoice price vs. no-arbitrage price?

$$\text{Invoice Price} = \text{Futures Price at Maturity} \times \text{Conversion Factor} + \text{Accrued Interest}$$

The no-arbitrage futures price is

$$\text{Futures Price at Maturity} = \frac{\text{Price of CTD Bond}}{\text{Conversion Factor of CTD Bond}}$$

How to find CTD bond?

1. Identify all available bonds falling within the maturity range (6.5–10 years)
2. Calculate the price of each one at the underlying's YTM (6%)
   - Conversion factor = PV bond CFs at 6% (expressed per dollar)
   - E.g. if the price at 6% is 105, the conversion factor is 1.05
3. The CTD bond is the bond that results in:

$$\text{Invoice Price} - \text{Futures Price at Maturity} \times \text{Conversion Factor} = 0$$

There should only be one bond that satisfies the above; all others will be negative

Describe repos.

Source: ILA101-113 Ch. 7: Interest Rate Forwards and Futures

- **Repo** – a sale of a security with a commitment to repurchase it at a specified price at a designated future date
  - The borrower is the one selling the security (receives cash for it)
  - The lender holds the security until the borrower repurchases it
- **Haircut** – the amount by which the repo collateral exceeds the loan

Clean vs. dirty price?

Dirty price = true market price of bond (PV bond CFs at YTM)

Clean price = Dirty Price $-$ Accrued Interest

Accrued interest = prorated portion of the coupon since the last coupon date

$$\text{Accrued Interest} = \frac{\text{Days Since Last Coupon}}{\text{Total Days Between Coupon Dates}}$$

Why are insurer duties complex?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Answering to policyholders, shareholders, rating agencies and tax agencies
- Managing its group’s overall balance sheet and capital adequacy
- Reporting financials based on accounting requirements (if publicly traded)

Insurer challenges in financial crisis?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Significant realized losses
- Unrealized losses via Accumulated Other Comprehensive Income (AOCI)
  - Many assets are classified as AFS $\rightarrow$ must reflect MV on balance sheet
  - AOCI = MV $-$ Amortized Cost
  - Flows through equity on the balance sheet
  - AOCI losses $\uparrow$ sharply between June 2008 and March 2009, then back to normal by September 2009
- Severe drops in market capitalization vs. the broader market
- Falling RBC ratios $\rightarrow$ need to raise capital

Insurer actions post-crisis?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

Increased liquidity, lowered risk

- Almost doubled allocations to cash and short-term securities
- Significantly reduced exposure to securitized assets

Raised capital

- Issued common and preferred equity
- Issued unsecured debt, surplus notes, and subordinated debt
- De-risked through changing asset allocations
- Attempted to divest businesses

List 3 ways to create integrated risk management strategy.

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

1. Clearly define a market risk budget
2. Evaluate economic objectives vs. insurance constraints
3. Determine how and where ALM fits into the overall risk management framework

List 6 core ERM processes.

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

1. Risk governance
2. Risk and capital measurement
3. Risk budgeting
4. Liquidity risk management
5. ALM and SAA
6. Risk reporting

List GSAM’s 6 ALM/SAA steps.

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Step 1 – Investment Objectives and Constraints
- Step 2 – Asset Universe and Assumptions
- Step 3 – Liability Cash Flow and Replicating Portfolio
- Step 4 – Risk Measures
- Step 5 – Risk Return Trade-Offs
- Step 6 – ALM and SAA

GSAM Step 1: Investment Objectives/Constraints?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

Gives insight into a portfolio's risk and return attributes under different scenarios

- Most important step

Examples of constraints:

- Achieve 5.5% target yield
- Asset-liability duration mismatch tolerance: ± 0.2
- Define exposure limits to specific asset/credit classes

GSAM Step 2: Asset Universe/Assumptions?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Establish a broad set of asset classes to achieve diversification and portfolio efficiency
- Correlations are key
  - Correlations between assets
  - Correlations between assets and liabilities

GSAM Step 3: Liability CF/Replicating Portfolio?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

3 main steps:

- Project liability cash flows to understand the liability profile
- Develop the duration profile of the liabilities
- Design a risk-minimizing asset portfolio that matches the key economic characteristics of the liabilities

GSAM Step 4: Risk Measures?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Helps assess effectiveness of different investment strategies
- Risk metrics:
  - Asset-only volatility
  - Surplus volatility
  - Economic or required capital, Value at Risk (VaR)
  - Conditional Value at Risk (CVaR) (a.k.a. CTE)
  - Surplus drawdown risk

GSAM Step 5: Risk/Return Trade-Offs?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Apply optimization techniques
- Evaluate and compare various efficient frontiers

GSAM Step 6: ALM/SAA?

Source: ILA101-112: Revisiting the Role of Insurance Company ALM

- Overall SAA decision reflects an insurer's risk and return objectives, ALM and other constraints
- SAA is used to develop customized benchmarks
- To recommend a portfolio, must also consider:
  - Appropriate benchmark(s) to be constructed in line with SAA
  - Relevant tolerance for active management to achieve target alpha
  - How will current market views be incorporated into the portfolio?

Capital market vs. actuarial ALM view?

Source: ILA101-114: ALM of Financial Institutions, Ch. 16

- **Capital Market View:** Insurance product = bundle of options granted by the insurer to policyholders in exchange for a fee
  - Preferred by the authors
  - Insurance products increasingly offer options that policyholders can exercise
- **Actuarial View:** Series of future benefit payments/cash flows tied to contingent events (death, etc.)
  - Relies on historical experience and current market for assumptions
  - If actual experience $\neq$ historical, insurers face substantial risk
  - Assumes economic variables stay within a reasonable range of historical experience

Optimal Solution = Robust ALM that combines:

- Actuarial approach (risk measurement, capital allocation)
- Capital markets view to manage embedded options

ALM experience loss examples?

Source: ILA101-114: ALM of Financial Institutions, Ch. 16

- **Guaranteed Investment Contracts (GICs):**
  - Falling interest rates make it difficult to fund guaranteed payments
- **Credit risk on high-yield and real estate**
  - Insurers were short credit options that were exercised (defaults)
- **VA GMxBs**
  - Unhedged insurers took losses when equity markets and rates $\downarrow$

Common Feature: Poor ALM practices susceptible to **paradigm market changes**

- Insurers wrote options based on an incorrect view of the world
- Products were mispriced, and portfolios overly risky

LTCI ALM lessons learned?

Source: ILA101-114: ALM of Financial Institutions, Ch. 16

- LTCI products may collect premiums decades before claims begin
- Pricing relies heavily on assumed reinvestment returns
  - Premiums must accumulate sufficient reserves
- Insurers in the 1990s assumed high reinvestment rates (7%)
- Unhedged insurers took large losses when rates ↓ in the early 2000s
- Erroneously assumed rates would mean revert higher

Lesson: Active management of options/rights protects against drops in future investment yield, increasing certainty of meeting obligations

Describe short options written by insurers.

Source: ILA101-114: ALM of Financial Institutions, Ch. 16

- **Risk Products:** Term life, disability, LTCI
  - **Feature:** Option to renew a policy at guaranteed premiums
  - **Embedded Short Option:** Call on the value of future benefit payments
- **Investment Products:** Fixed and variable annuities
  - **Features:** Additional deposits, withdrawals, guaranteed benefits/interest
  - **Embedded Short Options:**
    - Call on the value of future annuity payments
    - Put on the value of the policy
    - Interest rate floor
- **Hybrid Products:** WL and VUL
  - **Features:** cash surrender, borrowing at pre-determined rates
  - **Embedded Short Options:**
    - Put on the value of the policy
    - Series of puts on fixed-rate bonds

How can fixed annuity rate risk be hedged?

Insurer objectives:

- Crediting rate $\geqslant$ one-year LIBOR
- Earn a 150 bps spread (Yield $-$ Crediting Rate)

Assume: yield on 10-year investment = 6%, and LIBOR = 4.0%

- Objectives are met as long as LIBOR $\leqslant$ 4.5%
- If LIBOR rises above 4.5%, the insurer cannot earn a 150 bps spread

Solution: enter a swap: pay fixed (4.5%), receive floating (LIBOR)

- Locks in a spread of $6.0\% - 4.0\% - 4.5\% + 4.0\% = 150$ bps
- The floating LIBOR payment will always offset the crediting rate

Describe options granted to insurers.

Source: ILA101-114: ALM of Financial Institutions, Ch. 16

- **Risk Products:** Term life, disability, LTCI
  - **Features:** Right to receive either future premiums or reserve
  - **Right/option:** Callable bond
- **Investment Products:** Fixed and variable annuities
  - **Product features:**
    - Right to convert the policy from a fixed to floating rate
    - Right to receive future M&E fees and/or SCs (MV $\uparrow$ with underlying assets)
  - **Right/option:** swaption, bond with equity call option
- **Hybrid Products:** Permanent, whole, and variable life
  - **Product features:**
    - Right to change the dividend or NGE scale
    - Right to receive future premiums or excess of reserve over CSV
  - **Right/option:** swaption, callable bond